

Q1: 24 Feb (Shift 2) - Single Correct

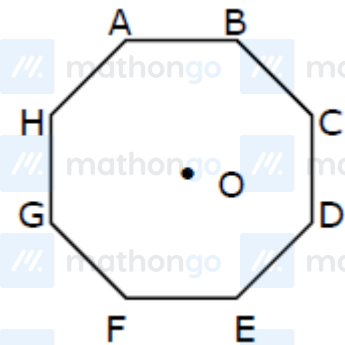
The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of 'L' is 1.0 m from meter scale having a minimum division of 1 mm and time of one complete oscillation is 1.95 s measured from stopwatch of 0.01 s resolution. The percentage error in the determination of 'g' will be :

- (1) 1.33%
- (2) 1.30%
- (3) 1.13%
- (4) 1.03%

Q2: 25 Feb (Shift 1) - Single Correct

In an octagon ABCDEFGH of equal side, what is the sum of $\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} + \vec{AG} + \vec{AH}$

If, $\vec{AO} = 2\hat{i} + 3\hat{j} - 4\hat{k}$



- (1) $16\hat{i} + 24\hat{j} - 32\hat{k}$
- (2) $-16\hat{i} - 24\hat{j} - 32\hat{k}$
- (3) $-16\hat{i} - 24\hat{j} + 32\hat{k}$
- (4) $-16\hat{i} + 24\hat{j} + 32\hat{k}$

Q3: 25 Feb (Shift 1) - Single Correct

The pitch of the screw gauge is 1 mm and there are 100 divisions on the circular scale. When nothing is put in between the jaws, the zero of the circular scale lines 8 divisions below the reference line. When a wire is

placed between the jaws, the first linear scale division is clearly visible while 72nd division on circular scale coincides with the reference line. The radius of the wire is :

- (1) 1.64 mm
- (2) 1.80 mm
- (3) 0.82 mm
- (4) 0.90 mm

Q4: 25 Feb (Shift 2) - Numerical

The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4% will be ____ %

Q5: 25 Feb (Shift 2) - Numerical

If $\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$, the angle between $\vec{P}$ and $\vec{Q}$ is θ ($0^\circ < \theta < 360^\circ$). The value of ' θ ' will be

Q6: 26 Feb (Shift 2) - Single Correct

A wire of 1Ω has a length of 1 m. It is stretched till its length increases by 25%. The percentage change in a resistance to the nearest integer is:

- (1) 25%
- (2) 12.5%
- (3) 76%
- (4) 56%

Answer Key**Q1 (3)****Q2 (1)****Q3 (3)****Q4 (2)****Q5 (180)****Q6 (4)**

Hints and Solutions

MathonGo

Q1 (3)

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T^2 = 4\pi^2 \left[\frac{\ell}{g} \right]$$

$$g = 4\pi^2 \left[\frac{\ell}{T^2} \right]$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$= \left[\frac{1 \text{ mm}}{1 \text{ m}} + \frac{2(10 \times 10^{-3})}{1.95} \right] \times 100$$

$$= 1.13\%$$

Q2 (1)

$$\overrightarrow{AO} + \overrightarrow{OB} = \overrightarrow{AB}$$

$$\overrightarrow{AO} + \overrightarrow{OC} = \overrightarrow{AC}$$

$$\overrightarrow{AO} + \overrightarrow{OD} = \overrightarrow{AD}$$

$$\overrightarrow{AO} + \overrightarrow{OE} = \overrightarrow{AE}$$

$$\overrightarrow{AO} + \overrightarrow{OF} = \overrightarrow{AF}$$

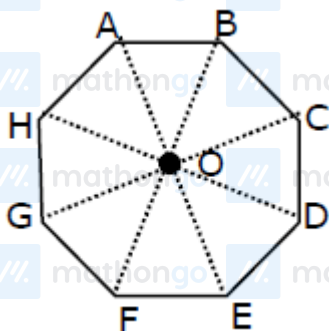
$$\overrightarrow{AO} + \overrightarrow{OG} = \overrightarrow{AG}$$

$$\overrightarrow{AO} + \overrightarrow{OH} = \overrightarrow{AH}$$

$$8\overrightarrow{AO} = (\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} + \overrightarrow{AG} + \overrightarrow{AH})$$

$$= 8(2\hat{i} + 3\hat{j} - 4\hat{k})$$

$$= 16\hat{i} + 24\hat{j} - 32\hat{k}$$



Q3 (3)

Hints and Solutions

MathonGo

$$\text{Least count} = \frac{\text{pitch}}{\text{no. of div.}} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$$

$$+ve \text{ error} = 8 \times \text{L. C.} = +0.08 \text{ mm}$$

$$\text{measured reading} = 1 \text{ mm} + 72 \times \text{L. C.}$$

$$= 1 \text{ mm} + 0.72 \text{ mm}$$

$$= 1.72 \text{ mm}$$

$$\text{True reading} = 1.72 - 0.08$$

$$= 1.64 \text{ mm}$$

$$\text{Radius} = \frac{1.64}{2} = 0.82 \text{ mm}$$

Q4 (2)

$$\text{Speed of transverse wave is } V = \sqrt{\frac{T}{\mu}}$$

$$\frac{\Delta n}{n} = \frac{1}{2} \ln T - \frac{1}{2} \ln \mu$$

$$\frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta T}{T}$$

$$= \frac{1}{2} \times 4$$

$$\frac{\Delta v}{v} = 2\%$$

Q5 (180)

$$\text{If } \vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$$

$$\text{Only if } \vec{P} = 0$$

$$\text{Or } \vec{Q} = 0$$

$$\text{The angle b/w } \vec{P} \text{ \& } \vec{Q} \text{ is } \theta (0^\circ < \theta < 360^\circ)$$

$$\text{So } \theta = 180^\circ$$

Q6 (4)

For stretched or compressed wire

$$R \propto I^2$$

$$\frac{R_1}{R_2} = \frac{l_1^2}{l_2^2}$$

$$\Rightarrow \frac{R}{R_2} = \frac{R^2}{(1.25l)^2}$$

% increase $\rightarrow 56.235\%$